A.R.J COLLEGE OF ENGINEERING AND TECHNOLOGY

PROJECT REPORT

TOPIC

EVENT FORCE MANAGEMENT SYSTEM

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

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ACADEMIC YEAR: 2026–2027

DECLARATION

We hereby declare that the project entitled “Event Force Management System” is our original academic work developed as part of the project requirements. The project focuses on using the Salesforce platform to organize event information, participant details, event activities, responsibilities and related records through a centralized management system.

The information presented in this report is based on the project design, configuration and testing carried out for the proposed system.

ACKNOWLEDGEMENT

We express our sincere gratitude to our institution, department and faculty members for providing the necessary facilities, guidance and encouragement to complete the Event Force Management System project.

We thank our project guide for valuable suggestions, continuous support and technical guidance throughout the development of the project.

We also thank our friends and teammates for their cooperation and support during the design, implementation and testing stages.

ABSTRACT

Event Force Management System is a Salesforce-based application designed to simplify the planning, coordination and monitoring of events. The system provides a centralized platform for maintaining event information, participant details, schedules, responsibilities, resources and event status.

The application uses Salesforce capabilities such as custom objects, fields, relationships, permissions, automation, Apex and Lightning Web Components. These features help organize event records and provide users with structured access to information.

The main purpose of the system is to reduce manual coordination, improve information accessibility, minimize data-entry errors and provide a systematic method for monitoring an event from planning to completion.

INTRODUCTION

Event management often involves multiple activities such as creating event records, assigning responsibilities, maintaining participant information, scheduling activities, coordinating resources and monitoring event progress. When these activities are handled through separate spreadsheets, messages or manual records, information can become difficult to maintain and retrieve.

The Event Force Management System uses Salesforce as a centralized cloud platform for event management. Users can create and maintain event records, manage participants, track event status and access related information through Salesforce interfaces.

Salesforce provides a scalable environment that supports structured data management, security, automation and customized interfaces for event-related operations.

PROBLEM STATEMENT

Managing events manually can involve several challenges such as:

- Difficulty in maintaining event records and schedules.
- Lack of centralized participant and event information.
- Manual updates of event status and activities.
- Difficulty in assigning and monitoring responsibilities.
- Increased possibility of data-entry and coordination errors.
- Limited visibility into event progress and pending activities.
- Difficulty in retrieving related event information quickly.

The Event Force Management System addresses these issues by providing a centralized Salesforce-based platform for managing event information and coordinating event-related activities.

OBJECTIVES OF THE PROJECT

1. To create a centralized event management system.
2. To maintain event, participant and activity information.
3. To track event status and progress.
4. To establish relationships between related event records.
5. To provide controlled and secure access to users.
6. To reduce manual data management and coordination.
7. To automate repetitive event-management processes.
8. To provide a simple and user-friendly interface using Salesforce Lightning components.

EXISTING SYSTEM

The existing event management process may depend on manual records, spreadsheets, separate communication channels or independent applications for maintaining event information.

Such approaches can make it difficult to maintain consistent data, coordinate responsibilities and obtain a complete view of event progress.

Common limitations include:

- Manual data entry and repeated updates.
- Difficult searching of event and participant records.
- Data duplication across different files.
- Limited automation of repetitive tasks.
- Difficult monitoring of event status.
- Less centralized information.
- Difficulty in tracking assigned responsibilities.

PROPOSED SYSTEM – EVENT FORCE MANAGEMENT SYSTEM

Event Force Management System is proposed as a Salesforce-based solution for managing event and coordination information. The system stores important information in Salesforce objects and allows authorized users to manage records through the Salesforce interface.

The proposed system provides:

- Centralized event records.
- Participant and organizer information management.
- Event schedule and activity tracking.
- Responsibility and task management.
- Object relationships between event-related records.
- Role and permission management.
- Automation for repetitive operations.
- Apex-based customized functionality.
- Lightning Web Component interface.

SALESFORCE ARCHITECTURE

Salesforce follows a cloud-based architecture in which application data, business logic, security and user-interface capabilities are managed through the Salesforce platform.

The Event Force Management System can be organized into the following layers:

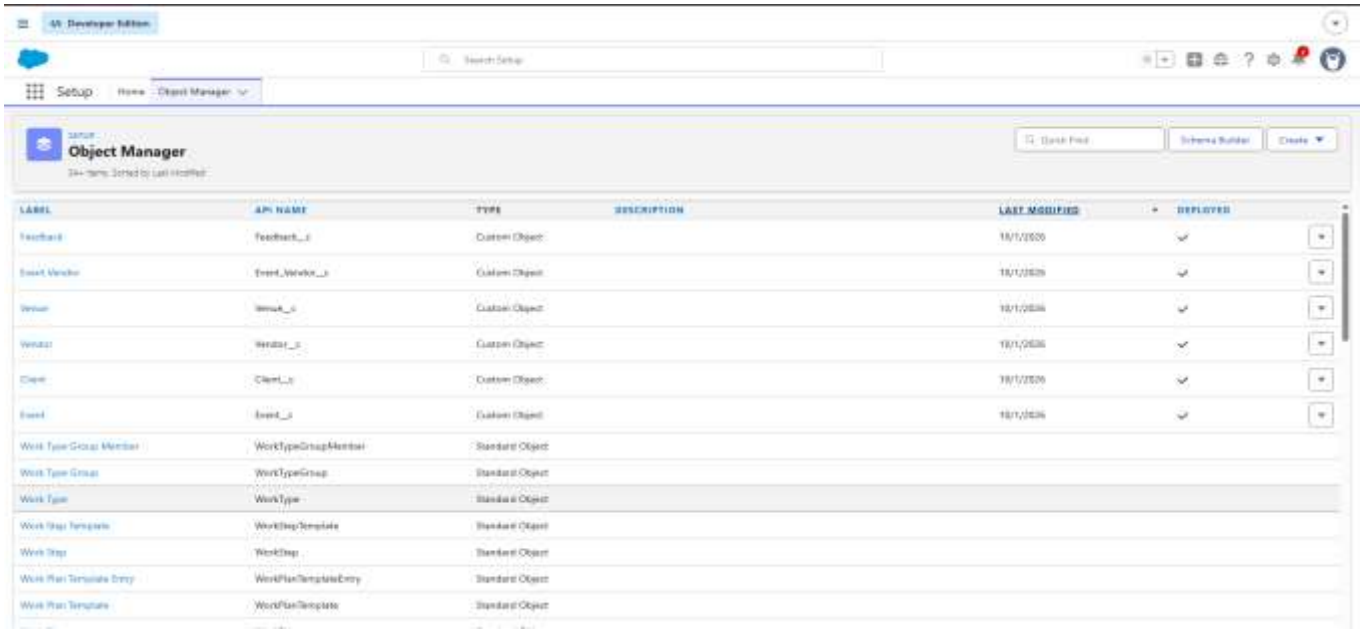
- Data Layer – Objects, fields and relationships used to store events, participants, activities and related information.
- Business Logic Layer – Apex and Salesforce automation used to implement event-management rules and repetitive processes.
- Security Layer – Profiles, permission sets, roles, object permissions, field-level security and sharing controls.
- Presentation Layer – Salesforce Lightning Experience and Lightning Web Components used to provide an interactive user interface.

This layered architecture helps maintain the application in a structured, secure and scalable manner.

OBJECTS AND FIELDS

Custom objects and fields are used to store Event Force Management System information. Objects represent the entities involved in event management, while fields store individual attributes.

- Event – Event name, date, venue, description, status and organizer.
- Participant – Participant name, contact information, registration details and participation status.
- Activity/Task – Activity description, assigned person, due date and completion status.
- Resource – Resource name, quantity, availability and allocation information.
- Organizer/Team – Responsible team or coordinator information.


Setup
Home
Object Manager
Object Manager
Quick Find
Schema Builder
Create
LABEL
API NAME
TYPE
DESCRIPTION
LAST MODIFIED
DEPLOYED
Work Type Group Member
WorkTypeGroupMember
Standard Object
Work Type Group
WorkTypeGroup
Standard Object
Work Type
WorkType
Standard Object

DATA MODEL

The data model defines how different records in the Event Force Management System are connected. Relationships can be used to associate an event with participants, activities, resources and responsible teams.

A typical relationship structure can be represented as:

EVENT → PARTICIPANTS

EVENT → ACTIVITIES / TASKS

EVENT → RESOURCES

EVENT → ORGANIZER / TEAM

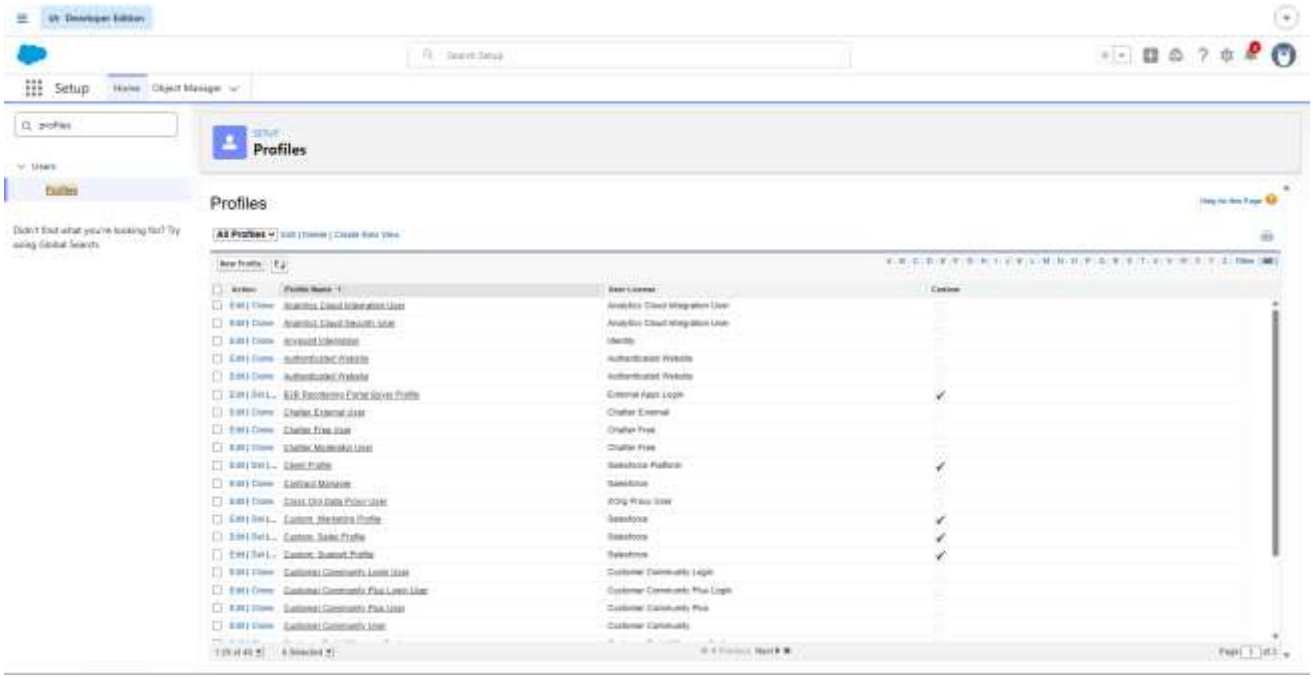
Using Salesforce relationships improves data organization and allows users to navigate between related records easily.

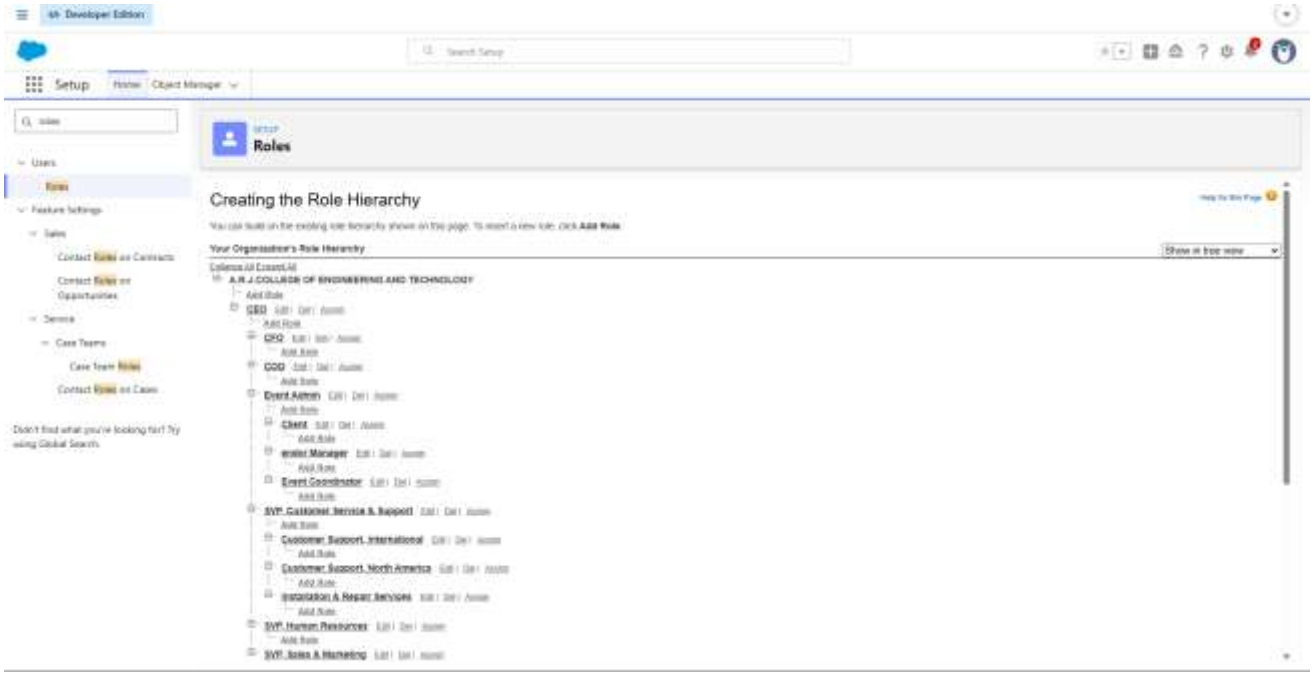
SECURITY AND ACCESS CONTROL

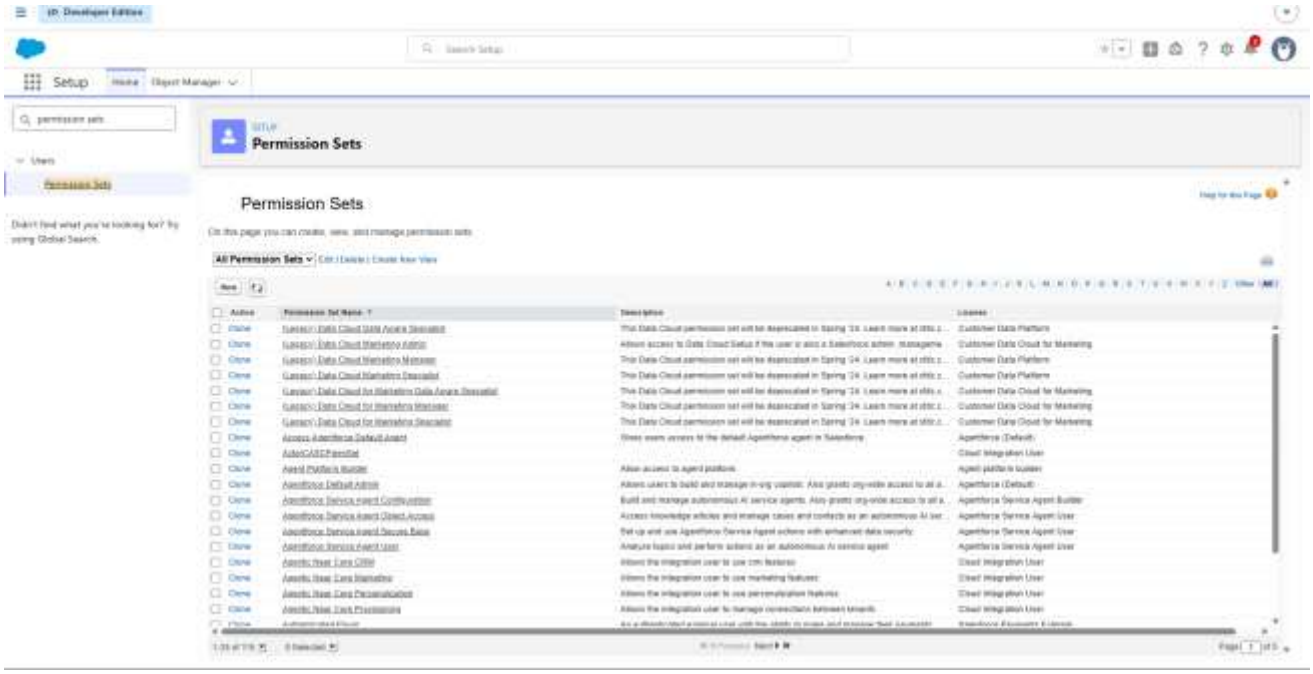
Salesforce security features are used to control access to Event Force Management System data.

- Profiles
- Permission Sets
- Roles
- Object Permissions
- Field-Level Security
- Sharing Settings

These controls help ensure that users can access only the information and operations required for their responsibilities.


Setup
Profiles
Profiles


Setup
Roles
Creating the Role Hierarchy

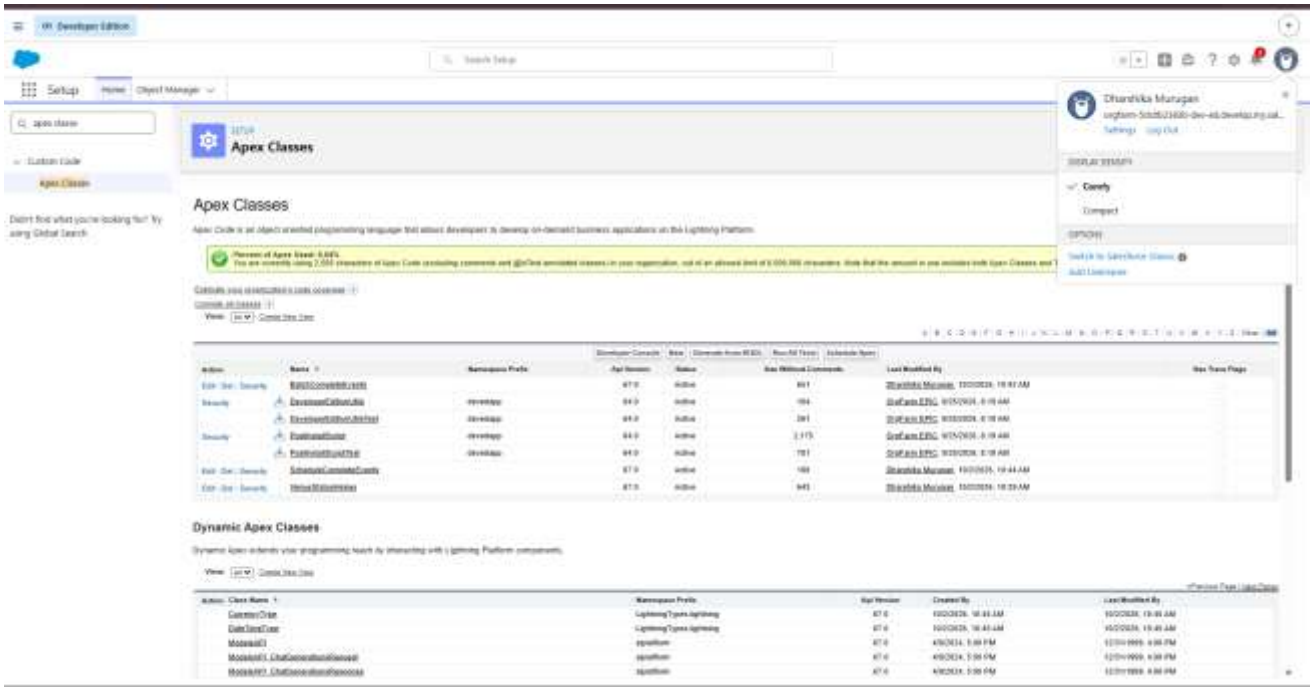

Setup
Permission Sets
Permission Sets

APEX AND AUTOMATION

Apex is Salesforce's programming language used to implement server-side business logic. In the Event Force Management System, Apex can be used for customized event-related processing that cannot be achieved through standard configuration alone.

Automation can be used to reduce repetitive manual activities and improve consistency. For example, automation may update event status, create follow-up tasks, validate required information or perform related record operations according to the configured business rules.

The use of Apex and automation provides flexibility for implementing project-specific functionality while keeping event records synchronized.

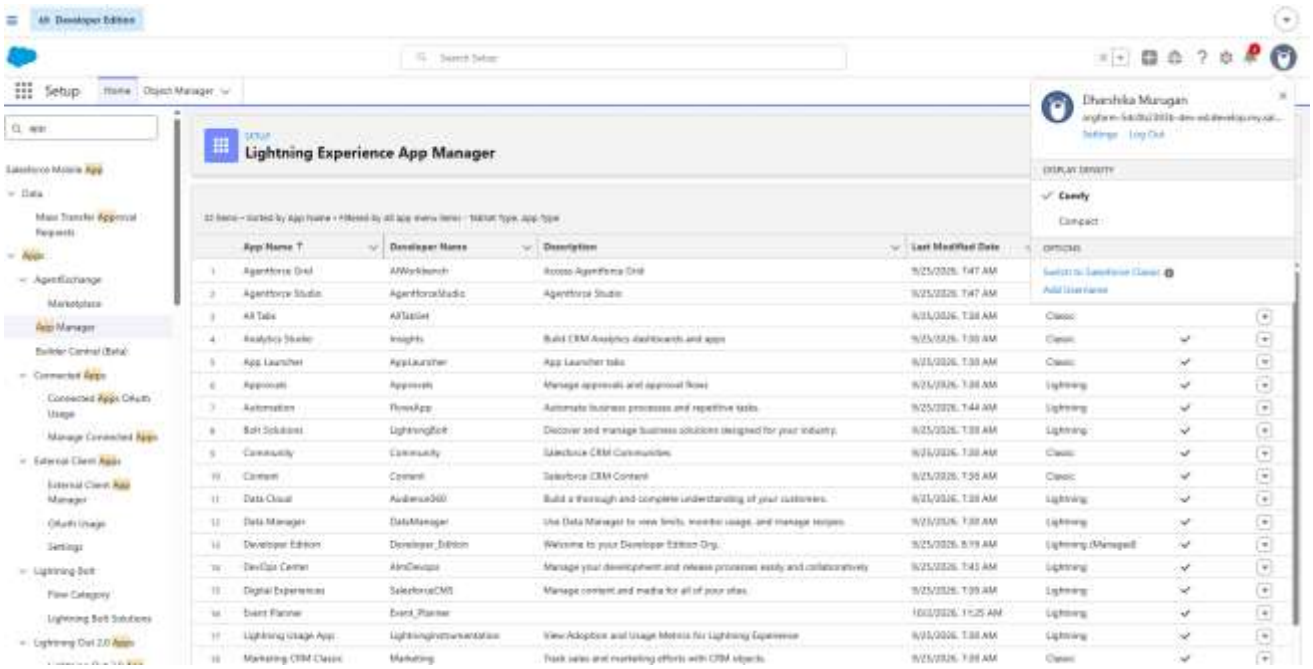


LIGHTNING WEB COMPONENTS

Lightning Web Components (LWC) can be used to create customized and interactive interfaces in Salesforce.

For the Event Force Management System, LWC can display event details, participant information, event status, activities and other related records in a user-friendly format.

Reusable components can interact with Salesforce data and provide users with a modern interface for event coordination.



SYSTEM IMPLEMENTATION

The Event Force Management System is implemented using Salesforce development and configuration features. The implementation process includes:

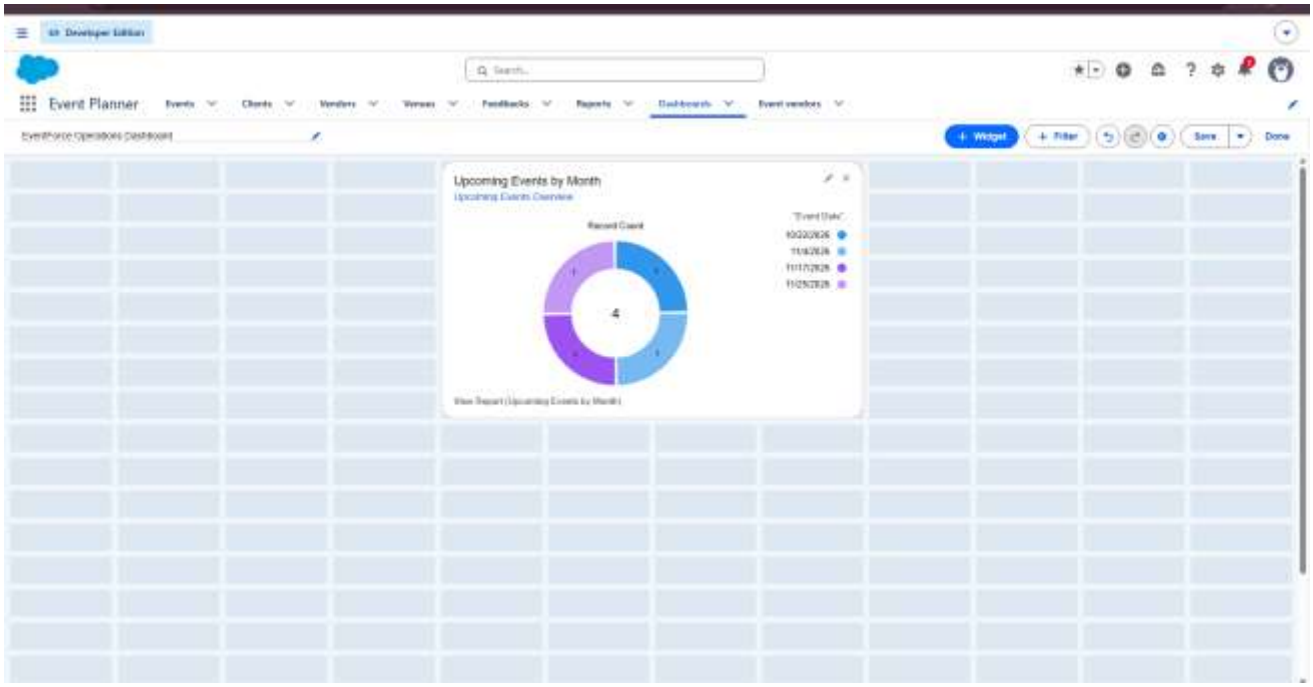
9. Creating Salesforce objects for event-related entities.
10. Creating required fields for each object.
11. Establishing relationships between related objects.
12. Configuring profiles and permissions.
13. Developing Apex functionality where required.
14. Creating Lightning Web Components for customized interfaces.
15. Configuring Salesforce automation.
16. Entering sample records and validating the workflow.
17. Testing the application against project requirements.

Each component should be tested individually and together to ensure that the complete system functions correctly.

FUNCTIONAL MODULES

- Event Registration and Record Management
- Participant Management
- Schedule and Activity Management

- Task/Responsibility Assignment
- Resource Management
- Event Status Monitoring
- User Access and Security
- Automation and Notifications

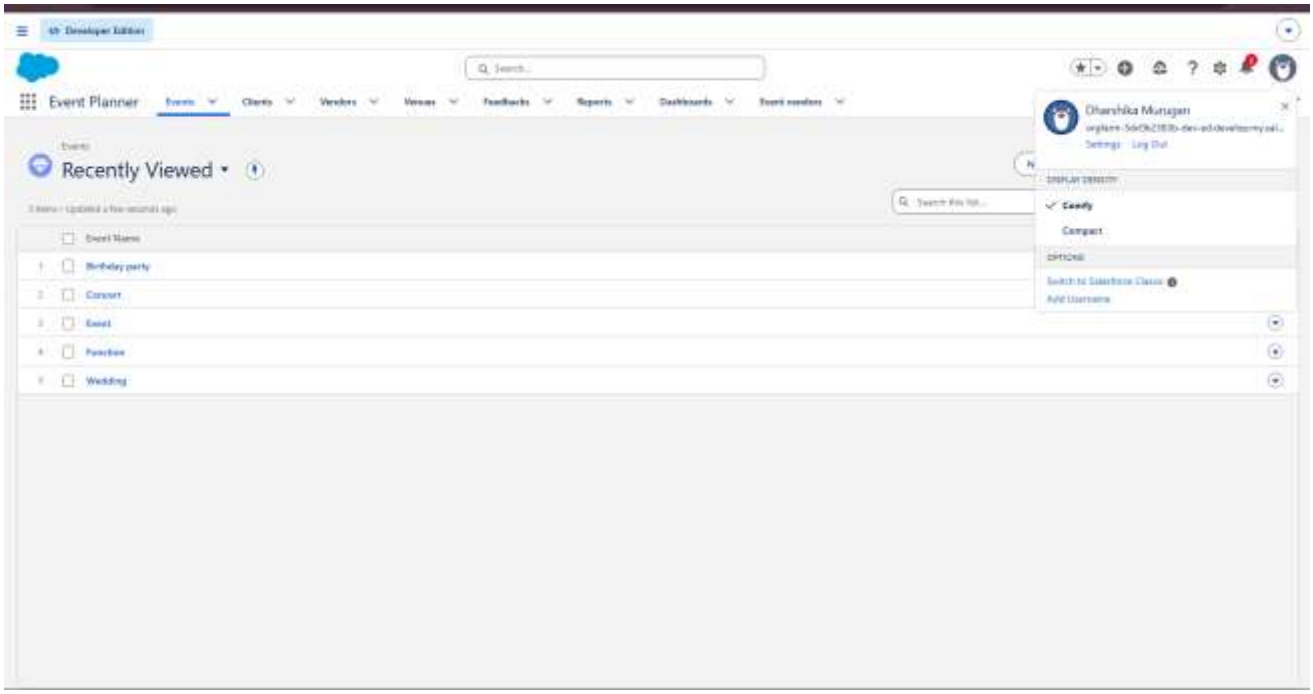


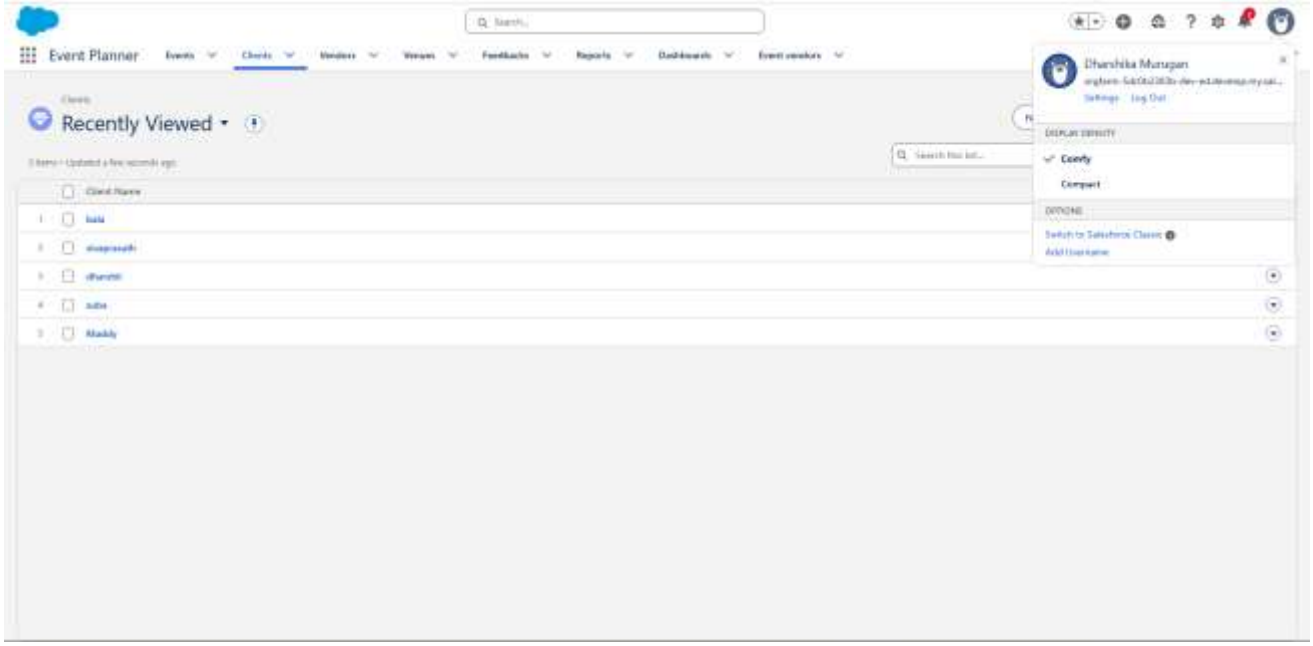
TESTING AND RESULTS

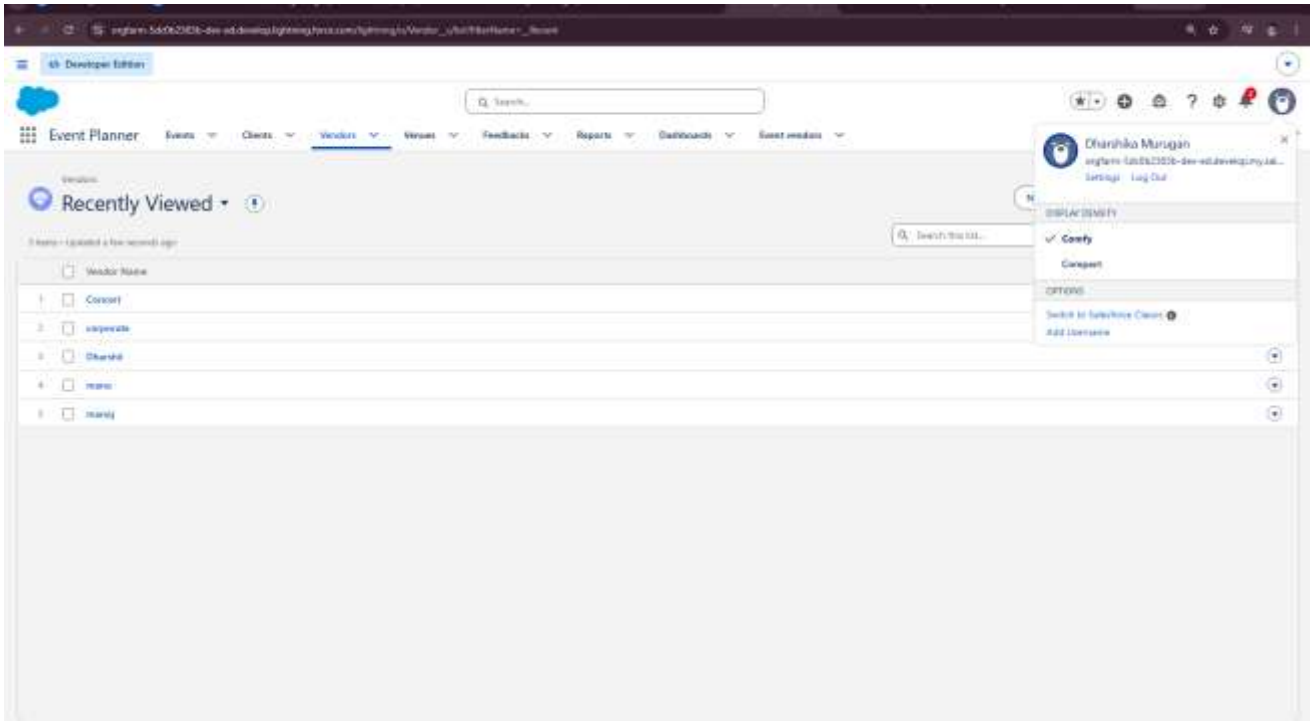
Testing is performed to verify the functionality of the Event Force Management System. The major areas to be tested include:

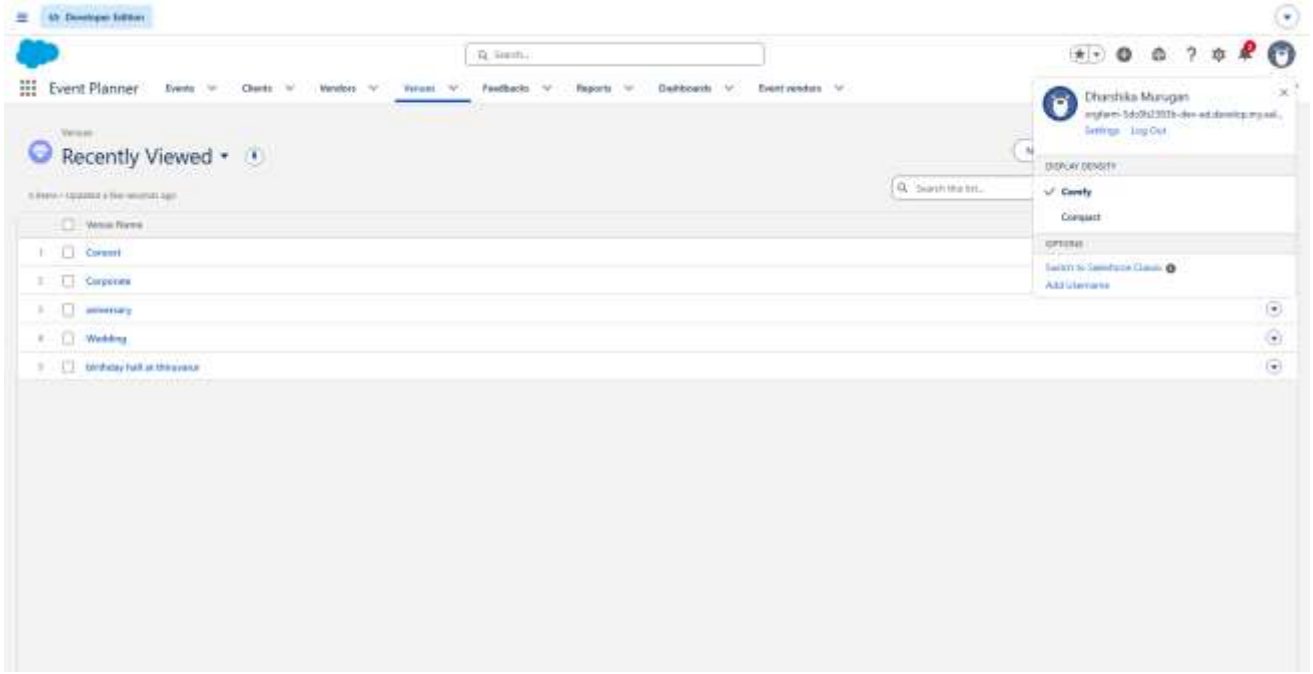
Test Area	Expected Result	Status
Event Record Creation	Event record should be created successfully	Pass
Field Validation	Correct and required values should be accepted	Pass
Participant Management	Participant information should be stored and retrieved	Pass
Event Status	Status should update correctly	Pass
Relationships	Related records should be accessible	Pass
Permissions	Users should access permitted data only	Pass
LWC	Component should display correctly	Pass
Apex	Required business logic should execute successfully	Pass
Automation	Configured automated action should execute correctly	Pass

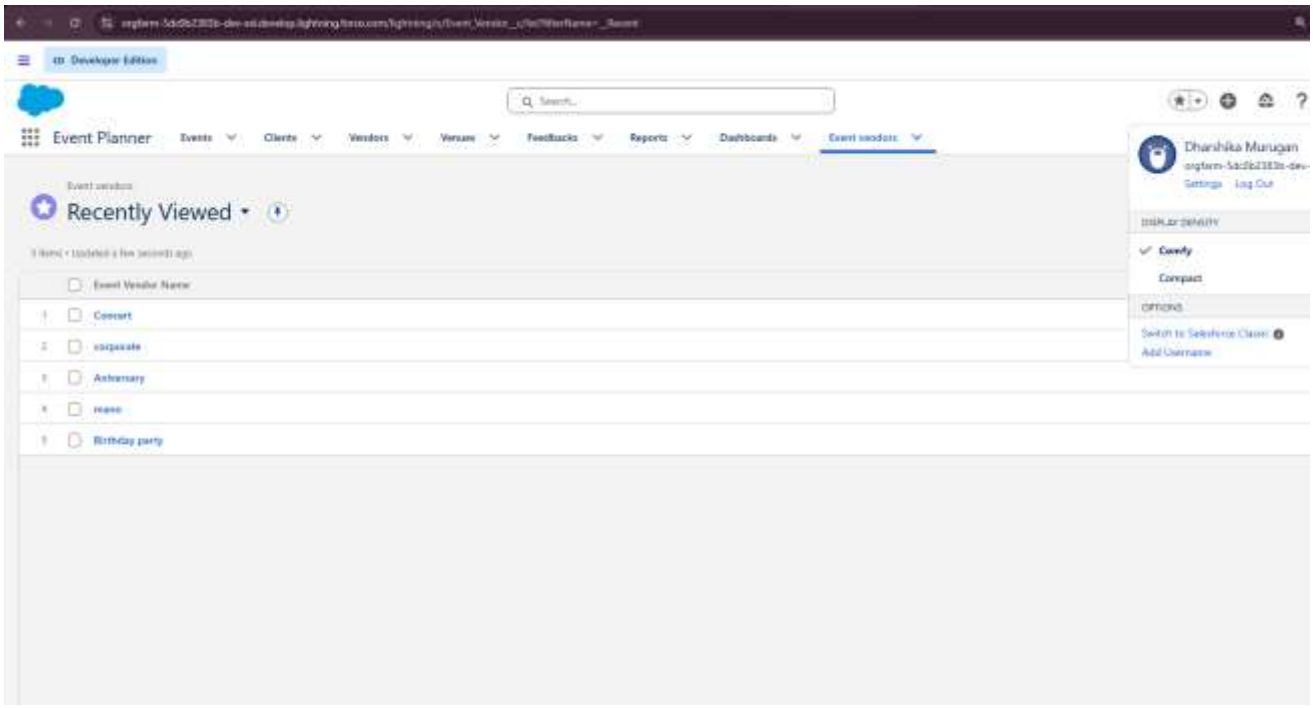
The testing process is intended to demonstrate that the configured Salesforce components work together to support event management.

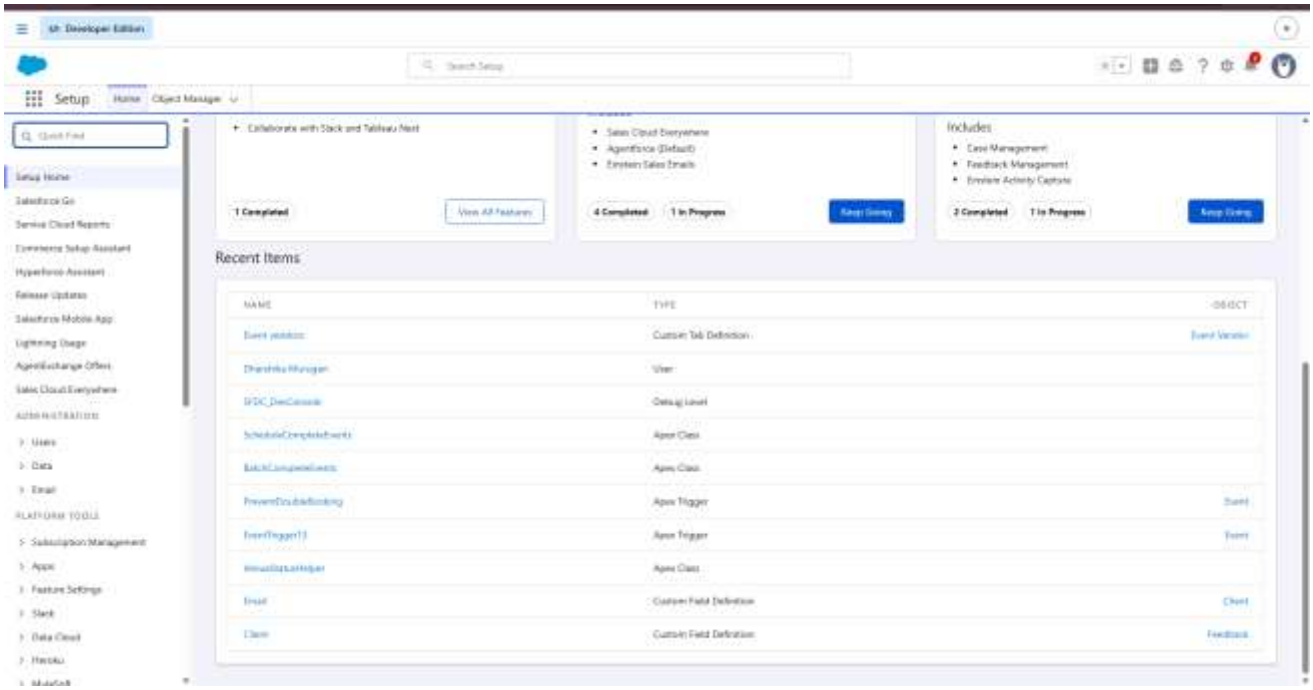

Event Planner
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Developer Edition
Event Planner
Events
Clients
Vendors
Venues
Feedbacks
Reports
Dashboards
Event vendors
Dharshika Murugan
Recently Viewed
Event Vendor Name
Concert
corporate
Anniversary
Birthday party


Developer Edition
Setup
Home
Object Manager
Recent Items
NAME
TYPE
OBJECT

ADVANTAGES

- Centralized event management.
- Easy access to event and participant information.
- Improved data organization.
- Secure and controlled user access.
- Reduced manual coordination.
- Customizable Salesforce interface.
- Automation of repetitive processes.
- Better visibility of event progress.
- Structured relationships between event records.

FUTURE ENHANCEMENTS

- Mobile application support for event coordinators.
- Automated email and SMS notifications.
- Advanced event analytics and dashboards.
- AI-assisted event planning and recommendations.
- Real-time attendance and check-in integration.
- QR-code-based participant check-in.
- Integration with external calendar services.
- Automated resource availability monitoring.
- Enhanced reporting for event performance.

CONCLUSION

- The Event Force Management System demonstrates how Salesforce can be used to develop a structured event planning, coordination and monitoring application.
- The project combines Salesforce configuration, data management, security, relationships, Apex, automation and Lightning Web Components to create a centralized platform for managing event information and related activities.
- The proposed implementation provides a foundation that can be further enhanced with mobile support, automated notifications, analytics, attendance integration and external service integrations.
- Overall, the system aims to reduce manual coordination, improve accessibility of event information and provide a more systematic approach to event management.

REFERENCES

- Salesforce Documentation.
- Salesforce Trailhead.
- Salesforce Developer Documentation.
- Apex Developer Guide.
- Lightning Web Components Documentation.

